

Lab 26: Case-Based Reasoning (Simple CBR System)

Aim

To implement a simple **Case-Based Reasoning (CBR) system** that retrieves the most similar past case and reuses its solution for a new problem.

Algorithm

1. Create a **case library** containing previous cases and their solutions.
2. Define a similarity function to compare the new case with stored cases.
3. Calculate the similarity score for each stored case.
4. Retrieve the most similar case.
5. Reuse the solution of the retrieved case for the new problem.
6. Retain the new case in the case library for future use.

Program

```
# Lab 26: Case-Based Reasoning (Simple CBR System)
```

```
# Case library containing previous cases
```

```
case_library = [  
    {  
        "symptoms": {  
            "fever": True,  
            "cough": True,  
            "fatigue": False  
        },  
        "diagnosis": "Flu"  
    },  
    {  
        "symptoms": {
```

```

        "fever": False,
        "cough": True,
        "fatigue": True
    },
    "diagnosis": "Cold"
},
{
    "symptoms": {
        "fever": True,
        "cough": False,
        "fatigue": True
    },
    "diagnosis": "Viral Infection"
}
]

```

Calculate similarity between two cases

```

def similarity(new_case, stored_case):
    return sum(
        1 for key in new_case
        if new_case[key] == stored_case.get(key)
    )

```

Retrieve the most similar case

```

def retrieve(new_case):
    scored_cases = [
        (
            similarity(new_case, case["symptoms"]),
            case
        )
        for case in case_library
    ]

```

```

]

return max(scored_cases, key=lambda x: x[0])[1]

# New problem
new_symptoms = {
    "fever": True,
    "cough": True,
    "fatigue": False
}

# Retrieve and reuse solution
best_case = retrieve(new_symptoms)

print("Retrieved Diagnosis (Reuse):",
      best_case["diagnosis"])

# Retain the new case
case_library.append({
    "symptoms": new_symptoms,
    "diagnosis": best_case["diagnosis"]
})

print("Case Library Size After Retain:",
      len(case_library))

```

Sample Output

Retrieved Diagnosis (Reuse): Flu

Case Library Size After Retain: 4

Result

Thus, the **Case-Based Reasoning system was successfully implemented**. The most similar previous case was retrieved, its solution was reused, and the new case was retained for future use.

